

Statement of Purpose

Program: Master of Science in Electrical Engineering (MSEE)

Applicant: Yang Sen (Liam) Lin

In my recent research accepted to **CoRL 2025**, titled “*See, Point, Fly*,” I identified a fundamental trade-off in the current state of robotic autonomy: geometric rules provide reliability but become brittle in dynamic environments; conversely, Generative Vision-Language-Action (VLA) models offer adaptability but suffer from hallucinations and high computational latency. My objective at Stanford is to resolve this tension by investigating the **mathematical interfaces** between rigid control theory and probabilistic foundation models. By integrating the safety-aware control frameworks of **Prof. Dorsa Sadigh’s ILIAD Lab** with the generalist robot learning of **Prof. Chelsea Finn’s IRIS Lab**, I aim to architect “Trustworthy Embodied Systems” that constrain the creativity of VLMs with the safety guarantees of control theory.

My journey toward this objective began with a focus on algorithmic decoupling. In my CoRL 2025 work, I formulated a geometric projection algorithm to ground 2D VLM prompts into 3D metric controls, solving depth ambiguity via pin-hole camera models without external sensors. By framing action prediction as a spatial grounding task rather than end-to-end text generation, I achieved state-of-the-art success rates (**93.9%**) in unstructured environments. However, through rigorous testing, I observed that my “learning-free” geometric priors—specifically the pin-hole camera model assumptions—became brittle when facing erratic, dynamic obstacles where non-linear reasoning is required. To overcome this “**reactivity bottleneck**” caused by VLM inference latency, I realized I must evolve from static heuristics to **Generalist Learning Policies**. I specifically require the curriculum at Stanford to investigate how meta-learning and stochastic control can replace static rules, enabling drones to adapt to non-linear dynamics in milliseconds.

Simultaneously, my research philosophy is defined by the constraints of the physical layer. While many researchers focus solely on algorithmic performance, my internship at **Wolley Inc.** grounded me in the realities of hardware architecture. There, I engineered C-based diagnostic tools to profile **CXL memory bandwidth** and host pressure. This experience gave me the low-level architectural intuition needed to optimize I/O efficiency for physical AI, distinguishing my approach as a full-stack system builder. I plan to extend this hardware-software co-design mindset during my upcoming internship at **OMRON SINIC X**, where I will distill Robotics Foundation Models for edge deployment. I recognized early on that algorithmic distillation is insufficient without optimizing the entire stack—from memory protocols to control loops—making the **Hardware/Software Systems** depth area at Stanford the ideal academic home for my work.

My academic preparation is defined by the convergence of **Mechanical Engineering** and **Computer Science**. While ME grounded me in the physical first principles of system dynamics, my dual focus in CS equipped me with the algorithmic reasoning of modern AI. I demonstrated the ability to synthesize these fields at a graduate level by securing an **A+ in Graduate Computer Vision**, proving my readiness for research-led curricula. Furthermore, I have actively applied advanced mathematics to solve engineering problems: I utilized **Projective Geometry** to derive non-linear unprojection laws in my *CoRL* paper and applied **Vector Calculus** to decompose displacement vectors into velocity primitives. This practical application prepares

me for the rigorous theoretical training of **EE 263 (Linear Dynamical Systems)** and **EE 364A (Convex Optimization)**. I specifically require **EE 364A** to formalize heuristic safety constraints into provably stable optimization layers, and **CS 330 (Meta-Learning)** to master the few-shot adaptation techniques necessary to move beyond my current zero-shot frameworks.

My technical roadmap directly maps to two specific research groups at Stanford. I aim to contribute to **Prof. Chelsea Finn’s IRIS Lab**, where I hope to apply my Sim2Real background to fine-tune generalist policies using large-scale interaction data, addressing the generalization limits I encountered in my CoRL work. Simultaneously, I seek to collaborate with **Prof. Dorsa Sadigh (ILIAD Lab)** to develop “safety filters”—similar to the geometric constraints I derived—that use control-theoretic guarantees to prevent VLM hallucination in safety-critical tasks.

Beyond the laboratory, I carry an “optimization mindset” from code to community. As the **GDSC Chapter Lead**, I directed 24 members to bridge the gap between academic theory and industry practice, organizing workshops that impacted over 40 student developers. I aim to bring this leadership to the Stanford community, fostering collaboration between EE and CS cohorts. My long-term goal is to lead the R&D of **Ubiquitous Physical AI**. The Stanford MSEE will transform me from a robotics engineer who builds working prototypes into a **Computational Systems Architect** capable of defining the safety guarantees required to deploy autonomous agents at a global scale.